

Assignment 2 – Comparative Life Cycle Assessment of Two Insulation Materials

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1. Introduction

This report uses Life Cycle Assessment (LCA) to determine the environmentally optimal external wall insulation type and thickness for the “Futura Haus neun” single-family house near Lund over 50 years, focusing on Ozone Depletion Potential (ODP) and Abiotic Depletion Potential of fossil resources (ADP fossil). The study uses LCA methods to assess two Swedish insulating materials in 50 mm increments from 0 to 500 mm thicknesses, integrating embodied impacts with operational energy effects via wall U value.

1.1 Concept definition

- **ODP – Ozone Depletion Potential**, is a midpoint indicator in LCA that measures the potential of substances like CFCs, halons, and HCFCs to deplete stratospheric ozone, expressed in kg CFC-11 equivalents. High ODP increases UV-B exposure risks to health and ecosystems. Mitigation includes using zero-ODP materials, ensuring containment, and promoting end-of-life recovery.
- **ADP fossil – Abiotic Depletion Potential, fossil (ADP fossil)**, is a midpoint LCA indicator quantifying fossil resource use (coal, oil, gas), typically in MJ or kg antimony equivalents. High ADP fossil reflects reliance on fossil energy and feedstocks across a product’s life cycle. Mitigation focuses on energy efficiency, low-carbon alternatives, material substitution, and circular design strategies.

2. Methodology

This project aims to analyze the quantitative environmental impact of the ODP/ADP-fossil emissions generated by the use of two different insulation materials, and to provide guidance in selecting one of the two materials.

To correlate this study to the current construction industry, mineral wool and XPS are chosen as they are arguably two of the most used products in today’s market in the Nordic region. (Ceresana)

The analysed insulation thickness is increased in steps of 50mm, starting from 0mm to 500mm, to establish the desired thickness.

2.1 Data selection

Products are selected based on the EPD files found on an international database. (International EPD System, n.d.) Considering that the transportation must be included in the LCA analysis local products are selected as their impact is presumably lower compared to a similar product from a far-away market. Therefore, a regional filter is set to Europe. (Eštoková & al)

2.2 EPD retrieved data

The following two products were selected from the database due to satisfying different criteria, such as providing all the stages considered relevant (A1-3, A5, C1, C2, C3, C4, D). The distance from the distribution centre to the construction site was also considered, although EPDs providing information on all stages were preferred:

- Mineral wool from Knauf. (Knauf Insulation, 2021)
- XPS from MASTERPLAST Nyrt. (Masterplast Nyrt, 2025)

Both EPDs came with a different functional unit, depending on the thickness required for a 1 m² panel to achieve an R-value of 1 m²*K/W. Considering the project's needs, the functional value was scaled in both instances for a 50mm thick 1m² panel and later increased in 50mm increments up to a total thickness of 500mm.

The original ODP values for both products are listed in the table below.

Table 1 ODP values retrieved from the EPD

Material	Unit	A1-A3	A5	C1	C2	C3	C4	D
Mineral wool (1m ² , R=3.05 m ² K/W)	kg CFC 11 eq.	9.76E-14	5.88E-11	0.00E+00	1.03E-17	0.00E+00	8.77E-16	-3.50E-13
XPS (1m ² , R=1 m ² K/W)	kg CFC 11 eq.	2,61E-08	4,33E-15	1,53E-13	1,37E-15	1,40E-13	3,01E-14	-9,73E-12

The original ADP-fossil values for both products are listed in the table below.

Table 2 ADP-fossil values retrieved from the EPD

Material	Unit	A1-A3	A5	C1	C2	C3	C4	D
Mineral wool (1m ² , R=3.05 m ² K/W)	MJ	2.77E+02	6.04E+00	0.00E+00	7.47E-01	0.00E+00	3.10E+00	-5.30E+00
XPS (1m ² , R=1 m ² K/W)	MJ	1,09E+02	1,18E-02	1,37E-01	1,48E-01	2,94E-01	1,46E-01	-1,83E+01

A4 values were manually calculated based on the distance and the quantity needed by selecting the values measured for an appropriate lorry, once the weight was determined based on the amount.

B1-7 stages were not considered, on the assumption that both insulations have an expected service life of equal to or greater than 50 years.

2.3 Energy, LCA calculations

By using an Excel spreadsheet template provided by the University, a basic energy calculation is performed representing the annual average heating energy need without any type of heat recovery.

The table below presents a few of the key parameters considered. All provided measurements are internal, with a 500mm layer of insulation. Any variation to the insulation layer results in an increase in the internal measurements.

Table 3 Key energy parameters

Height	12,5 m
Width	6,5 m
Length	7 m
λ – value (XPS)	0,034 (W/mK)
λ – value (mineral wool)	0,039 (W/mK)
Average outdoor temperature (Lund)	8,4 °C

All energy results for heating without any heat recovery are later correlated with the given ADP/ODP values based on another University template and added to the total given by the material itself.

Resulted material values dependent on the insulation thickness are added up for the whole analysed LCA stages, resulting in an ODP and ADP total for the material's full life-stage. By multiplying with the considered façade area and then dividing with the heated floor area, the result per $1 \text{ m}^2 A_{\text{Temp}}$ is achieved. Therefore, the functional units of this analysis are: kg CFC 11 eq./ $1 \text{ m}^2 A_{\text{Temp}}$ year, respectively MJ/ $1 \text{ m}^2 A_{\text{Temp}}$ year, considering a lifespan of 50 years.

2.4 Considering the future

As the planet warms, temperatures are expected to rise globally. (NASA, 2022) Therefore, the average outdoor temperature used in the energy calculations is not expected to remain constant. When constructing a new building, a 50-year scenario is considered; in such cases, the future must be taken into account and the construction adapted accordingly.

Inspired by the Paris Agreement, two different scenarios are analyzed, where the average outdoor temperature increases by 2 °C to 10 °C (+1.6 °C compared to the present) and 11 °C (+2.6 °C compared to the present). (United Nations, 2015)

3. Results

Results in this section include figures with all final ODP and ADP numbers. Moreover, the sensitivity analysis, or a parametric study, of the global warming potential in terms of energy consumption for heating is concluded as “Energy +” at 1.6 °C and 2.6 °C in all figures below.

3.1. ODP – Mineral Wool

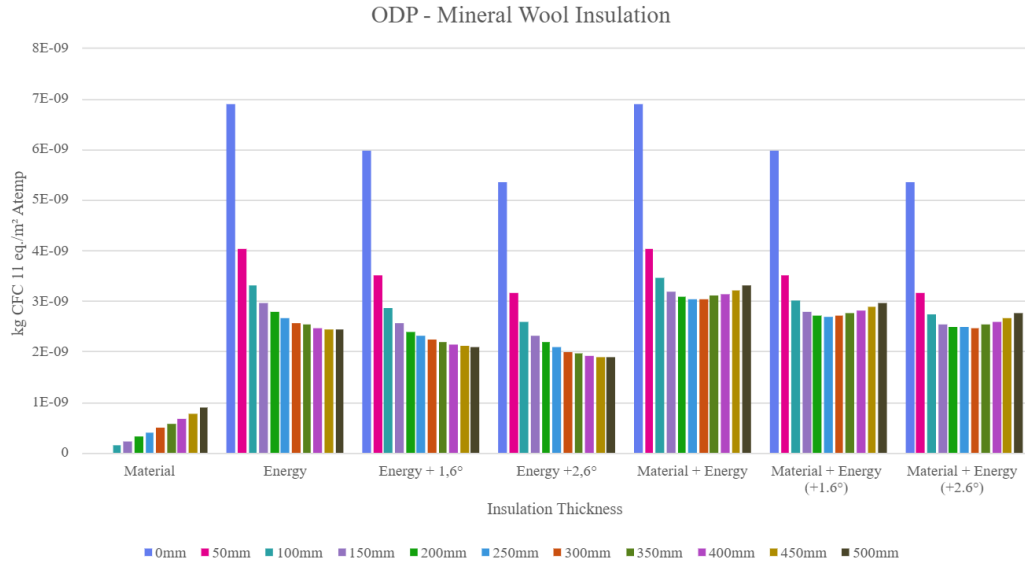


Fig. 1- Ozone Depletion Potential (ODP) results of different mineral wool thicknesses

The heating energy of the building clearly dominates ODP for the analysed mineral wool. The material, mineral wool, located at the far left of the figure, does not significantly contribute to ozone depletion potential compared to the energy required at today's temperature levels, nor in the future, given higher average temperatures due to climate change. A balance between energy and material ODP can be achieved by choosing 200-300mm of Insulation itself.

3.2. ODP – Expanded Polystyrene

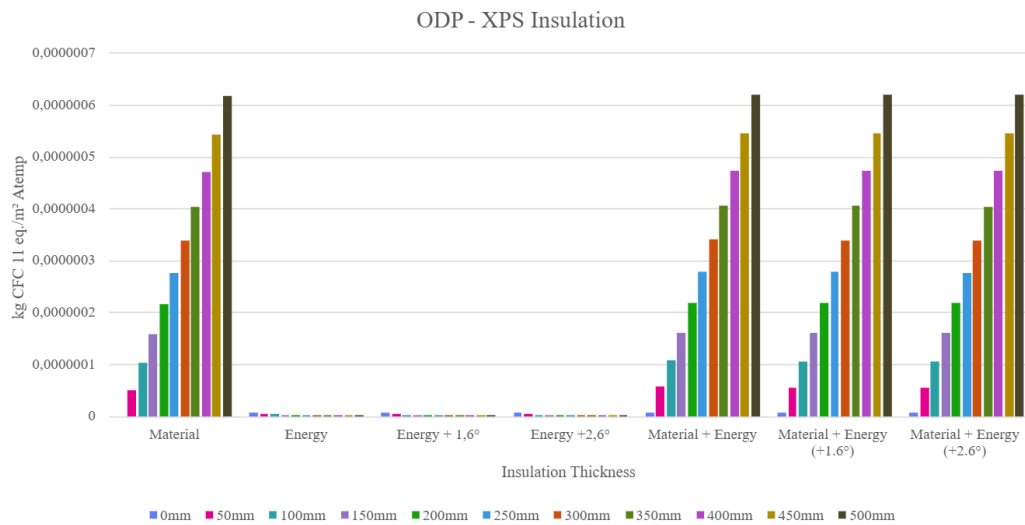


Fig. 2- Ozone Depletion Potential (ODP) results of different XPS thicknesses

On the other hand, when it comes to XPS insulation, it clearly dominates the ODP chart, making it harder to conclude which option is better or worse. Energy does not play a significant role in this figure, as the ODP potential is only a fraction of what XPS contributes during all analyzed LCA stages throughout its lifetime.

3.3. ADP (Fossil) – Mineral Wool

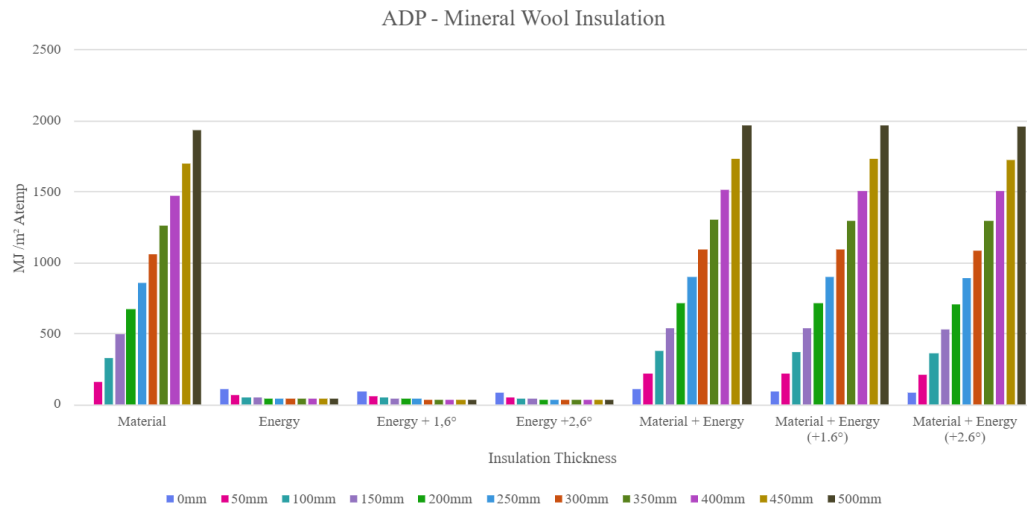


Fig. 3- Abiotic Depletion for fossil resources (ADP-Fossil) results of different mineral wool thicknesses

The same phenomenon that occurs with ODP for expanded polystyrene also happens with ADP mineral wool. The material itself largely dominates the figure compared to the energy consumed for heating the building, and follows a largely linear path as the thickness of the material increases.

3.4. ADP (Fossil) – Expanded Polystyrene

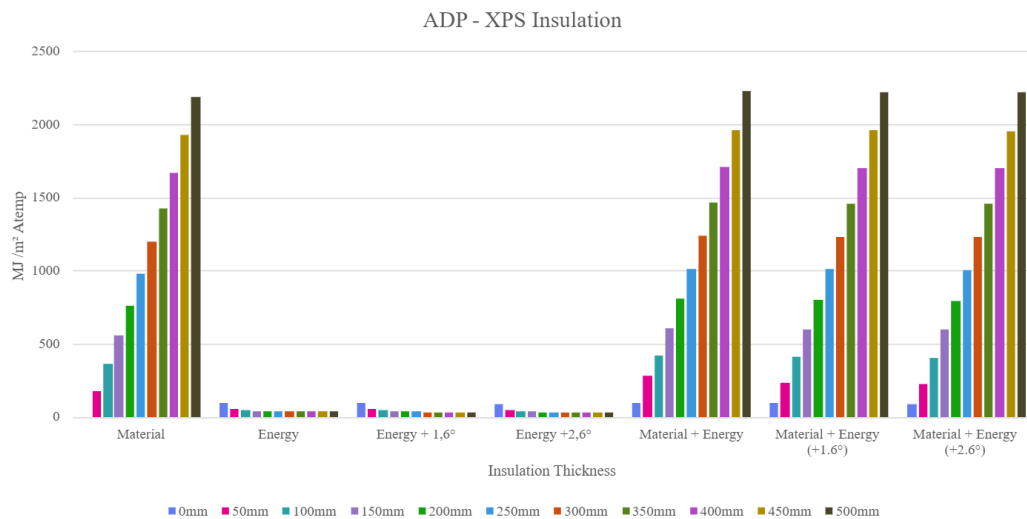


Fig. 4- Abiotic Depletion for fossil resources (ADP-Fossil) results of different XPS thicknesses

ADP for XPS insulation again follows a linear path upward as the material thickness increases, while dwarfing the heating energy consumption for both current and future temperature increases.

4. Discussion

Mineral wool exhibits a reduced material impact compared to XPS and appears to influence energy efficiency more significantly, providing a greater range of thickness options. Mineral wool with a thickness ranging from 200mm to 350mm may provide the optimal balance without compromise. Nonetheless, the lifetime Ozone Depletion Potential (ODP) of XPS surpasses its energy savings. The choice between energy and materials is untenable, as materials consistently exert the greatest influence on Ozone Depletion Potential (ODP). As the significance of heating energy use diminishes, sensitivity analysis indicates that increasing temperatures will necessitate a greater focus on selecting less harmful materials. A direct comparison of the analyzed materials indicates that mineral wool exhibits greater sustainability for Ozone Depletion Potential (ODP) and Abiotic Depletion Potential (ADP) than extruded polystyrene (XPS), which possesses an excessively high ODP even at reduced thicknesses, thereby casting doubt on its historical, present, and future applications.

5. Conclusion

This study indicates that mineral wool surpasses XPS in terms of Ozone Depletion Potential (ODP) and fossil Abiotic Depletion Potential (ADP). The impact of mineral wool escalates with thickness, albeit marginally, whereas XPS incurs severe material-related burdens in both categories. Other LCA studies have determined that 200–350 mm of mineral wool equilibrates embodied and operational impacts.

The findings must be interpreted with prudence. Only two impact categories were evaluated, service life assumptions were simplified, and B-stages were omitted. These selections constrain the scope of sustainability assessment. Anticipated warming scenarios diminish the significance of heating energy, amplifying material considerations in long-term design (Sohn, 2017).

In summary, mineral wool is the more suitable choice. For successful decision making, it is important to include supplementary categories (e.g., global warming potential, human health effects, recyclability, cost) and to consider data quality prior to generalizing results.

6. References

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Appendix

Short Form

LCC
LCA
NPV
ODP
ADP Fossil
EDP

Abbreviations

Life Cycle Cost
Life Cycle Assessment
Net Present Value
Ozone Depletion Potential
Abiotic Depletion Potential, Fossil
Environmental Product Declaration